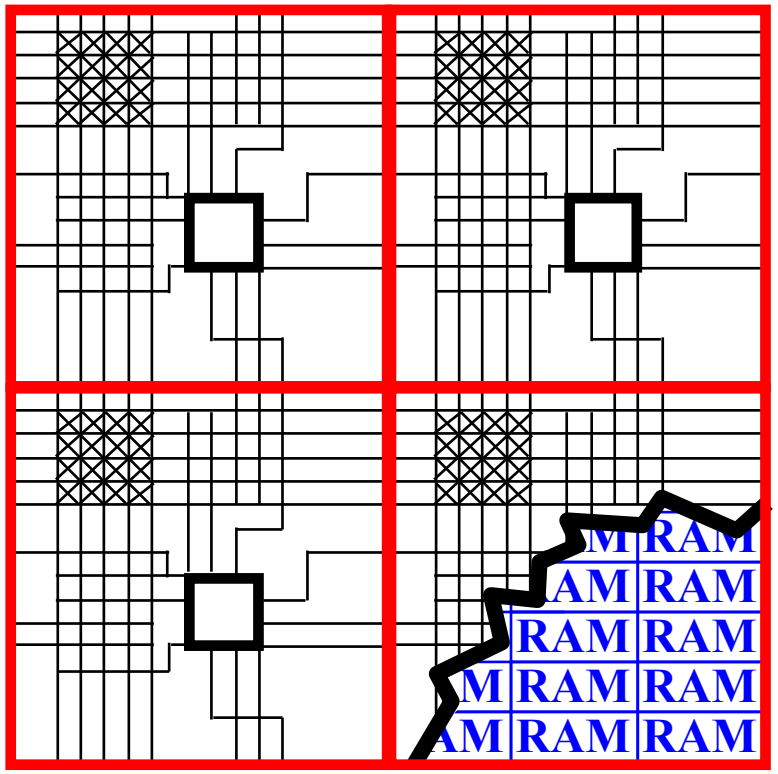
LiDAR Car 4.6

Spring quarter!

New leads!

- Each subteam will have a lead to guide them through activities
- We are here to help when you get stuck and teach skills, not be hands-on for you
- Please ask leads questions if you don’t understand *why* something is being done

Field Programmable Gate Arrays (FPGAs)



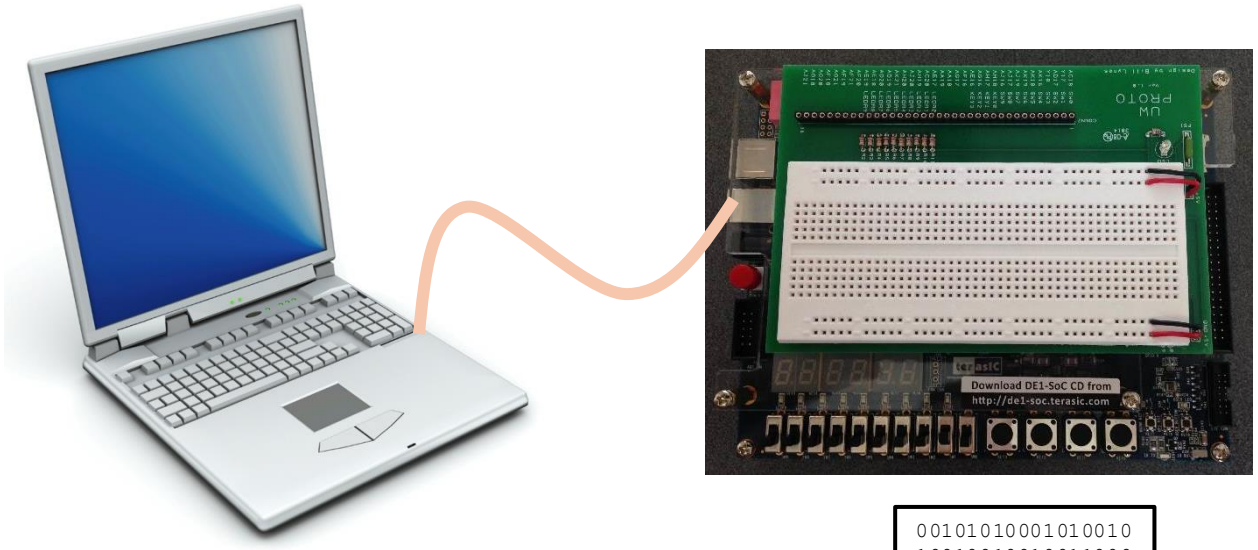
Logic cells imbedded in a general routing structure

Logic cells usually contain:

- **6-input Boolean function calculator**
- **Flip-flop (1-bit memory)**

All features electronically (re)programmable

Using an FPGA



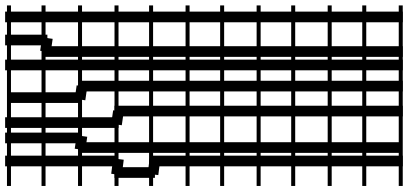

```
//  Verilog  code  for  2-input
multiplexer

module AOI (F, A, B, C, D);
  output F;
  input A, B, C, D;

  assign  F  =  ~((A  &  B)  |  (C  &
D));
endmodule

module MUX2 (V, SEL, I, J);    //
2:1 multiplexer
  output V;
  input SEL, I, J;
  wire SELB, VB;

  not G1 (SELB, SEL);
  AOI G2 (VB, I, SEL, SELB, J);
  not G3 (V, VB);
endmodule
```

Verilog

FPGA
CAD
Tools

```
00101010001010010
10010010010011000
10101000101011000
10101001010010101
00010110001001010
10101001111001001
01000010101001010
10010010000101010
10100101010010100
01010110101001010
01010010100101001
```

Bitstream



Simulation

Verilog

- Programming language for describing hardware
  - Simulate behavior before (wasting time) implementing
  - Find bugs early
  - Enable tools to automatically create implementation
- Similar to C/C++/Java
  - VHDL similar to ADA
- Modern version is “System Verilog”
  - Superset of previous; cleaner and more efficient

deliverables

Sensors:

1. Read raw incoming packets from lidar into computer

2. FPGA -> Arduino communication (verilog basics)

Controls:

1. Wire four motors to motor driver and control them simultaneously

2. CAD design (putting components into onshape)
    - github/demos/parts_list

Closing thoughts

- Someone from each team (not the lead) say what you accomplished today and what you want to do next week
- Bring cardboard if you can!